

EXP2-2 – Twists and Turns

The **aim** of EXP2-2 is to consider how to plan a set of measurements to determine the shear modulus of materials from the twisting of a wire.

The **objectives** for this exercise are:

- Plan a set of experimental measurements
- Study the simple harmonic motion of a torsion pendulum and use it to calculate the shear modulus of a metal and polymer wire and damping factor of a polymer wire
- Consider the uncertainty of measurement in the results for the shear modulus
- Critically discuss the results

To do this we will need to use what we have learned about measurement and planning experiments from the last exercises and from QXU4007 and QXU4011 in the previous years.

Background information:

The shear modulus, G , is the material elastic constant in shear loading:

$$\text{Shear Modulus} = \frac{\text{Shear Stress}}{\text{Shear Strain}}$$

Shear modulus is related to the Young's modulus, E , and the Poisson's ratio, ν , by equation 1,

$$G = \frac{E}{2(1+\nu)} \quad (1)$$

When a wire is twisted (torsion) it is being loaded in pure shear. We can find the shear modulus by making the twisting wire oscillate in simple harmonic motion. From the time period of the oscillation the shear modulus can be determined. This is called a **torsional pendulum**.

A mass in the form of a disc is suspended by a wire whose shear modulus is to be measured (Figure 1).

By letting the mass freely oscillate the shear modulus, G , can be found from equation 2,

$$G = \frac{4\pi L m R^2}{r^4 T^2} (Pa) \quad (2)$$

where L is the length of the wire and r is the wire radius, m is the mass of a disc with radius R , and T is the period of oscillation (time for 1 complete cycle).

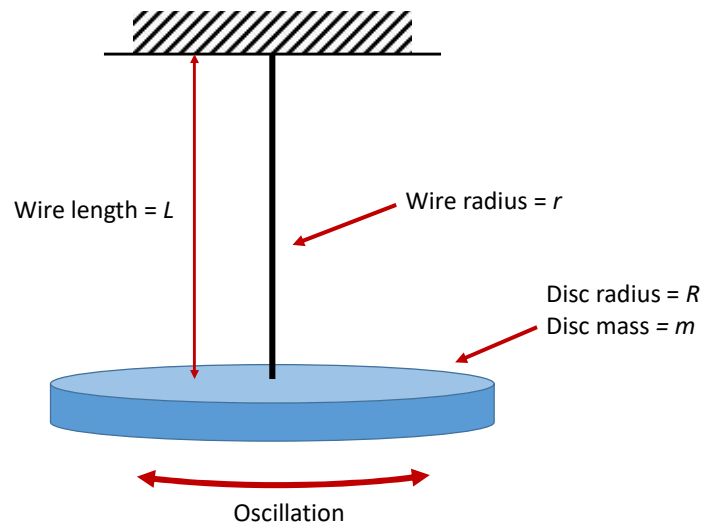


Figure 1. Torsional pendulum

The shear modulus is a constant for a material (an elastic response in which energy is conserved). It can be seen from Equation 2 that the shear modulus of the wire could be determined from a single measurement of the torsional pendulum. However, it is better scientific practice to make multiple measurements that change the contribution of uncertainties in the measurement. By rearranging equation 2 we can see that plotting T^2 as a function of L gives the constant factors in the experiment as the slope, from which the shear modulus, G , can be found.

The time period of the oscillation does not depend on the amplitude of the oscillation, it only depends on the length of the wire (this is the basis of the pendulum in a clock). However, the oscillation amplitude will decrease with time due to energy losses in the system (Figure 2).

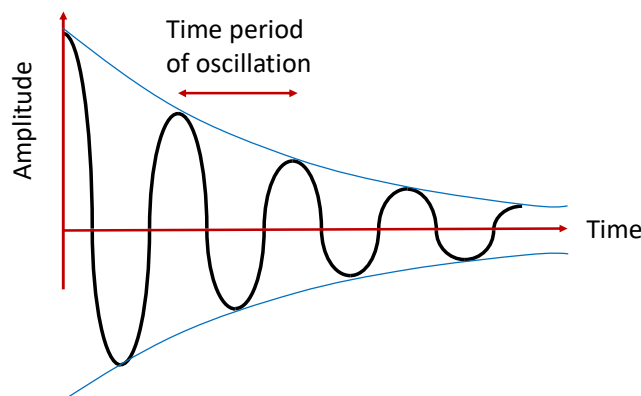


Figure 2. Amplitude decays over time but the period of oscillation remains constant.

The reduction in amplitude of the oscillation (mechanical damping) comes from 2 sources – external and internal. The external sources of damping are due to friction between the moving object and the air. These are minimised by the disc shape of the mass. The internal sources of damping come from absorption of energy in the material of the wire. For the metal wire, below the elastic limit, the internal damping is negligible. For the polymer wire the internal damping is high. For this reason, a metal bell rings for a long time but a plastic bell does not.

The logarithmic decrement, κ , can be found from the ratio of amplitudes, A_0/A_n , over n cycles and a plot of $\ln \frac{A_0}{A_n}$ vs n gives κ as the slope. The amount of damping can be characterised by:

$$\text{The quality factor, } Q = \frac{\pi}{\kappa} \quad (3)$$

$$\tan \delta = \frac{1}{Q} \quad (4)$$

What you should do:

Before conducting the experiments, it is necessary to plan ahead as to what you would do in the laboratory. In your groups, read the descriptions of the experiment given above and consider how you would make the measurement to determine the shear modulus of a metal and a polymer wire.

You should consider:

- How will you determine G ?
- What experiments are necessary and what are the variable quantities?
- What values need to be measured?
- Which parameter(s) will contribute the largest uncertainties?
- What will you use to measure each value?
- How precisely does the measurement need to be?
- How many times do you need to repeat the measurements?

You can assume that:

- The disc is 100mm in diameter and weighs 1kg
- The metal wire is 0.6mm (Group 1&2) or 0.6096mm in diameter
- The polymer wire is 1mm in diameter

The elastic constants of the materials are:

	Young's modulus / GPa	Poisson's ratio
Metal wire	210	0.30
Polymer wire	1.7	0.40

From these consider the time period of oscillation for different lengths of wire and **estimate** the results. What will your graphs and calculations look like?

Write a plan for the experimental measurements.

Briefly describe the **aims of your experimental** plan. What is the principle of the experiment? How will the shear modulus of the 2 wires be determined? You do not need to repeat the methods given in this document; you can give the reference to the documents on QMPlus.

From the assumed values for the material elastic constants and parameters of the experiment, estimate the values of L and T that you would use to determine the shear modulus. Plot these estimated values on a **graph** to show what the experiment might look like.

Make a **table of all the parameters** that need to be measured. For each parameter:

- give the approximate measurement value(s)
- the type of instrument you would measure them with
- the reading resolution of each instrument
- the approximate precision of the measurement
(this should be aligned with the approximate measured values)
- how many times each measurement should be taken

The experimental plan will be **no more than 600 words excluding titles, names and reference**.

You may use narrow page margins and letters no smaller than font size 11.

The **title** of the plan will be: **EXP2-2 Experimental Plan XX** (where XX is your group number, e.g. M16 or P8). The names of **all the group members** should be given as authors of the experimental plan unless someone showed NO contribution to the group work.

You will **submit** your group's plan as a **pdf document on QMPlus** using the file name:

'QXU5017 EXP2-2 Part 1 Plan XX.pdf' (where XX is your group number, e.g. M9 or P19).

The plan must be completed and **submitted by 24:00 on Friday 14th March 2025**.

The plan must be the **work of the whole group** and the content agreed by all group members.

Only one group member need to hit the submit button to submit the report.

You will submit your **'Risk Assessment Form – Twists and Turns XX'** (where XX is your group number).

The form must be completed, submitted and **approved (pass) by 24:00 on Tuesday 18th March 2025**.

Only one group member needs to hit the submit button to submit the report.

Marks breakdown:

EXP2-2 is 30% of the marks for QXU5017

Part 1 Group Plan 1 will be worth 40% of EXP2-2

Part 2 Group Report 2 (Instructions will be given in Week 5) will be worth 60% of EXP2-2

Schedule:**M Groups**

Date:	Time:	Place:	Activity:
Monday 10 th	10:30-12:10	A310	Feedback to EXP2-1 Introduction to EXP2-2
Wednesday 12 th	10:30-12:10		Group Meeting 1 Formulate experimental plan
Friday 14 th	08:30-10:10		Group Meeting 2 Complete and Submit plan
Complete and submit Risk assessment form until you pass			
Wednesday 19 th	10:30-12:10	A108	Data collection (M1-M10)
Friday 21 st	08:30-10:10		Data collection (M11-M20)
Week 6			
Monday 24 th	10:30-12:10	A310	Group Meeting 3 (Define uncertainty)
Wednesday 26 th	10:30-12:10		Group Meeting 4 (Write report)
Friday 28 th	08:30-10:10		Submit report

P Groups

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